

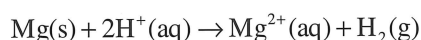
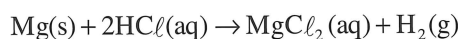
Set 36: Acid and base reaction stoichiometry

Notes

Example (mass to volume calculation)

A student carried out a laboratory experiment similar to Experiment 23, hoping to produce enough hydrogen gas to fill a 2.00 L balloon at S.T.P. The student reacted 3.00 g of magnesium in excess, 2.0 mol L⁻¹ hydrochloric acid solution. Will enough hydrogen be produced?

- (a) Write an equation for the reaction



- (b) Calculate the number of moles of magnesium

$$n(\text{Mg}) = \frac{m}{M} = \frac{3.00}{24.3} = 0.123 \text{ mol}$$

- (c) Calculate the number of moles of hydrogen gas produced

$$n(\text{H}_2) = n(\text{Mg}) = 0.123 \text{ mol}$$

- (d) Calculate the volume of hydrogen gas produced (1 mole of a gas occupies 22.71 L at STP)

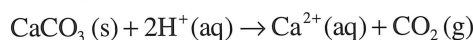
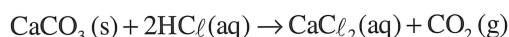
$$V(\text{H}_2) = n(\text{H}_2) \times 22.71 = (0.123) \times 22.71 = 2.79 \text{ L}$$

The reaction will produce enough gas to fill the 2.00 L balloon.

Example (volume to moles and mass calculation)

Brick cleaning acid reacts with limestone, calcium carbonate, producing carbon dioxide gas. How many moles and what mass of limestone is needed to produce 2.00 L of carbon dioxide gas at STP?

- (a) Write an equation for the reaction



- (b) Calculate the number of moles of carbon dioxide gas (1 mole of a gas occupies 22.71 L at S.T.P.)

$$n(\text{CO}_2) = \frac{V(\text{CO}_2)}{22.71} = \frac{2}{22.71} = 0.0881 \text{ mol}$$

- (c) Calculate the number of moles of calcium carbonate needed

$$n(\text{CaCO}_3) = n(\text{CO}_2) = 0.0881 \text{ mol}$$

- (d) Calculate the mass of calcium carbonate needed

$$m(\text{CaCO}_3) = n(\text{CaCO}_3) \times M(\text{CaCO}_3) = 0.0881 \times 100.09 = 8.81 \text{ g}$$

Set 36: Exercises

1. A 20.0 mL sample of stomach juices obtained from a patient suffering from hyperacidity was neutralised using 10.4 mg of solid aluminium hydroxide. Assuming that the volume of stomach juice present in the patient's stomach at the time was 250 mL and the only acid present was hydrochloric acid, calculate the mass of acid present in the stomach.
2. Sweat is a solution containing mainly sodium, potassium and chloride ions. During vigorous exercise, particularly in hot and humid conditions this solution often runs off the skin. In doing so it dissolves significant amounts of hydrogen ions from the skin. The concentration of hydrogen ions in sweat is typically $1.00 \times 10^{-5} \text{ g L}^{-1}$. Calculate the mass of solid calcium hydroxide required to neutralise 10.0 mL of sweat.
3. A 2.89 g sample of sandstone, containing only calcium carbonate and silicon dioxide, is analysed by reacting it with hydrochloric acid. 10.7 mL of 2.50 mol L^{-1} hydrochloric acid solution is required for complete reaction. Calculate the percentage by mass of calcium carbonate in the sandstone sample.
4. Hydrochloric acid is used to decrease the pH of swimming pool water. To determine the amount of acid to be added, the concentration of the acid needs to be known. To determine the concentration of some hydrochloric acid, 5.60 g of pure calcium carbonate in the form of calcite was placed into 20.0 mL of the acid. When the reaction was completed the remaining calcite is removed, washed in distilled water and dried. It was found to have a mass of 5.09 g. Calculate the concentration of the hydrochloric acid.
5. To determine the purity of limestone, mostly calcium carbonate, found in a cave, an analyst measured the mass of a small sample of limestone to be 2.59 g. He placed enough hydrochloric acid into a beaker to completely dissolve the limestone sample. The beaker and acid had a mass of 110.61 g. He then placed the limestone sample into the acid and allowed it to completely dissolve. After the reaction was completed the mass of the beaker and its contents was found to be 112.22 g. Calculate the percentage purity of the limestone.
6. Sodium carbonate can be used to neutralise acid spills in the environment. It can be used as a solid or as a solution that can gradually seep into the soil.
 - (a) Calculate the concentration, in mol L^{-1} , of a sodium carbonate solution produced from $2.50 \times 10^2 \text{ g}$ of sodium carbonate-10-water dissolved into 2.00 L of water to produce 2.10 L of solution.
 - (b) If the acid spilt was hydrochloric acid, calculate the mass of sodium chloride produced in the soil if all of the 2.10 L of the sodium carbonate solution reacted with acid in the soil.
7. Slaked lime (Ca(OH)_2) is sometimes used to reduce soil acidity. To neutralise the top 200 mm of soil in a paddock required $1.00 \times 10^2 \text{ g}$ of slaked lime per square metre. Calculate the number of moles of hydrogen ions per square metre in the top $2.00 \times 10^2 \text{ mm}$ of soil.
8. $9.00 \times 10^4 \text{ L}$ of rain water with an average hydrogen ion concentration of $1.00 \times 10^{-4} \text{ g L}^{-1}$ was collected and stored in a galvanised iron tank. After several months the pH of the water was found to be 7.00 ($[\text{H}^+] = 1.00 \times 10^{-7} \text{ mol L}^{-1}$). Assuming the zinc was the only material in contact with the water, calculate the
 - (a) mass of zinc that would have dissolved
 - (b) the concentration of zinc ion, in mol L^{-1} , in the rain water.



Question 4: Hydrochloric acid

Notes